Let us analyse the effect of the addition of 0.01 mol of solid sodium hydroxide to one litre of a buffer solution containing 0.8 M CH_3COOH and 0.8 M CH_3COONa. Assume that the volume change due to the addition of NaOH is negligible. (Given: K_a for CH_3COOH is 1.8×10^{-5})

$$\underset{0.8-\alpha}{CH_3\text{-}COOH(aq)} \overset{H_2O}{\rightleftharpoons} \underset{\alpha}{CH_3COO^-(aq)} + \underset{\alpha}{H^+(aq)}$$

$$\underset{0.8}{CH_3COONa(aq)} \xrightarrow{H_2O} \underset{0.8}{CH_3COO^-(aq)} + \underset{0.8}{Na^+(aq)}$$

The dissociation constant for CH_3COOH is given by

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]};$$

$$[H^+] = K_a \frac{[CH_3COOH]}{[CH_3COO^-]}$$

The above expression shows that the concentration of H^+ is directly proportional to $\frac{[CH_3COOH]}{[CH_3COO^-]}$.

Let the degree of dissociation of CH_3COOH be α then,

$[CH_3COOH]=0.8-\alpha$ and $[CH_3COO^-]=\alpha+0.8$

$$\therefore [H^+] = K_a \frac{(0.8-\alpha)}{(0.8+\alpha)}$$

$\alpha << 0.8$, $\quad \therefore 0.8-\alpha \simeq 0.8$ and $0.8+\alpha \simeq 0.8$

$$[H^+] = \frac{K_a(0.8)}{(0.8)} \Rightarrow [H^+] = K_a$$

Given that

K_a for CH_3COOH is 1.8×10^{-5}

$\therefore [H^+] = 1.8\times10^{-5}$; pH = $-\log(1.8\times10^{-5})$

$= 5 - \log 1.8$

$= 5 - 0.26$

pH $= 4.74$

Calculation of pH after adding 0.01 mol NaOH to 1 litre of buffer.

Given that the volume change due to the addition of NaOH is negligible $\therefore [OH^-] = 0.01M$. The consumption of OH^- are expressed by the following equations.

$$\underset{0.8-\alpha}{CH_3COOH\,(aq)} \rightleftharpoons \underset{\alpha}{CH_3COO^-(aq)} + \underset{\alpha}{H^+(aq)}$$

$$\underset{0.8}{CH_3COONa(aq)} \rightarrow \underset{0.8}{CH_3COO^-(aq)} + \underset{0.8}{Na^+(aq)}$$

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